Om Bharati

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Summary

Computer Science student specializing in Python backend architecture and local AI infrastructure. Focused on building resource-efficient systems using FastAPI, Docker, and open-weight LLMs.

Education

B.Tech in Computer Science **Expected graduation: 2030**

Gandhi Institute for Education and Technology, Baniatangi, Bhubaneswar, Odisha

Projects

Spreadsheet Engine Python | GitHub

- Built a spreadsheet engine with cell storage, formula evaluation, and JSON and CSV persistence.
- Used Python's `ast` module to safely parse and evaluate cell formulas.
- Engineered a dependency-driven recalculation system using Directed Acyclic Graphs (DAGs) to detect circular references and optimize updates.

CYBERGUARD Hackathon team project | GitHub

- Architected a local multi-model LLM pipeline orchestrating a decision model (Laya) and a reasoning model (Qwen 8B), heavily optimized to run efficiently within a strict 6 GB VRAM constraint.
- Contributed to the architecture and DevOps of a threat-intelligence platform whose URL analyzer screens URLs locally against a dataset of 2M+ records.
- Designed a risk-analysis workflow that separates structured risk decisions from LLM-generated explanations.

Technical Skills

Language: Python

Backend: FastAPI, REST APIs

Database: PostgreSQL, Redis, pgvector

AI & Inference: LLM APIs, RAG (Retrieval-Augmented Generation), Local Model Orchestration

Workflow & Infrastructure: Linux, Git, Docker, GitHub Copilot, Google Antigravity IDE (AI-assisted scaffolding)